

JALDRISHTI

AI + GIS Watershed Intelligence & Field Evidence Platform

Smart India Hackathon 2026 • SIH26015

Application of Geospatial Techniques for visualization and analysis to interpret Geo-Coded Images to enhance watershed Development Outcomes

Working decision

Proceed with SIH26015. The problem aligns strongly with a map-first, evidence-first web prototype and can be demonstrated convincingly within a short build window. The solution should NOT be a generic GIS dashboard; it should center on intervention-level before/after evidence, geo-coded field photos, outcome indicators and explainable assessment.

Competition snapshot

An indexed SIH 2026 viewer captured on 2 Oct 2026 showed 178/500 ideas and a 5 Oct 2026 submission deadline. Your stated ~180/500 is therefore consistent with the indexed snapshot. Because other public mirrors show conflicting/stale counters and dates, the live SIH portal should be checked immediately before submission; do not cite a stale mirror as the official count.

1. Executive Summary

JALDRISHTI is proposed as an operational watershed intelligence workbench for the Ministry of Rural Development / Department of Land Resources use case. It combines geo-coded field photographs, watershed boundaries, intervention/work records, satellite-derived indicators and terrain context into one auditable map-based workflow. The primary user is not a casual map viewer; it is a district/state officer who needs to answer: What intervention happened? Where did it happen? What changed after it? What evidence supports that conclusion? What needs field verification next?

The strongest differentiator is an evidence chain rather than a larger dashboard: each intervention becomes a spatial evidence bundle containing the field image, GPS/time metadata, nearby satellite time series, watershed context, terrain indicators, comparison window, outcome dimensions, confidence, and reviewer status. The platform explicitly avoids claiming causality when the data can only support association or an observational change.

Element	JALDRISHTI decision
Problem framing	Turn geo-tagged documentation into analysable evidence for watershed monitoring and outcome assessment.
Demo focus	One watershed → one intervention → before/after evidence → outcome assessment → field verification action.
Primary UI	Map-first command center with intervention markers, time slider, evidence drawer and outcome scorecard.
Core analytics	NDVI, MNDWI/NDWI, NDMI, LST where valid, terrain/slope/flow context, change detection, anomaly flags.
AI role	Explain patterns, prioritize reviews and summarize evidence; not a black-box causality engine.
Data strategy	Use public satellite/terrain datasets for demo; design

	adapters for SRISHTI-DRISHTI and DoLR MIS/geo-coded data.
Fast-build principle	Frontend-first deterministic demo data, then optional live backend/analytics adapters.
Judge story	Evidence → change → confidence → action, not “another GIS dashboard”.

2. Verified SIH26015 Challenge Context

The indexed problem statement identifies the organization as the Ministry of Rural Development and the department as the Department of Land Resources (DoLR). The statement is software-oriented and sits in the Agriculture, FoodTech & Rural Development theme. Its background explicitly calls out the need to integrate geo-coded imagery with satellite-based spatial datasets and to improve planning, monitoring, interpretation and scientific assessment of watershed interventions.

The problem statement specifically names a 30 m satellite-data opportunity through the SRISHTI-DRISHTI platform and asks for thematic/analytical outputs involving land degradation, water conservation structures, vegetation change, drainage conditions and other watershed characteristics. It also highlights the under-utilization of geo-tagged photographs beyond basic documentation.

Verified item	Current working fact	Source
PS ID	SIH26015	Indexed SIH 2026 PS viewer
Title	Application of Geospatial Techniques for visualization and analysis to interpret Geo-Coded Images to enhance watershed Development Outcomes.	Indexed SIH 2026 PS viewer / problem statement mirrors
Organization	Ministry of Rural Development	Indexed SIH 2026 PS viewer
Department	Department of Land Resources (DoLR)	Indexed SIH 2026 PS viewer
Category / theme	Software / Agriculture, FoodTech & Rural Development	Indexed SIH 2026 PS viewer
Current indexed submissions	178/500 in a 2 Oct 2026 capture; use “~180/500” for planning	sih2026-ps-viewer capture
Working deadline	5 Oct 2026 in the same capture	sih2026-ps-viewer capture
Dataset pointer	Google Drive link is listed with the PS	Problem statement viewer

Deadline caution

One community-maintained archive shows a 30 Sep deadline and another mirror shows much smaller counters. Because the 2 Oct snapshot specifically records 178/500 and 5 Oct, and this matches the user-provided situation, the build plan uses 5 Oct. Verify the official SIH portal at submission time.

3. Deep Problem Research

3.1 The operational gap

Watershed programs already collect structured MIS information, field observations and spatial/geo-tagged information. The missing layer is an integrated evidence workflow that joins those records with remote-sensing change and shows an officer what changed at the exact intervention location, with traceable provenance.

3.2 Documentation is not outcome evidence

A photograph proves that an image was captured at a location/time; it does not by itself prove ecological or hydrological improvement. Satellite trends can show spatial change, but they do not automatically prove the

intervention caused that change. JALDRISHTI therefore separates evidence types and displays confidence/limitations.

3.3 Fragmentation

Official WDC-PMKSY systems expose reports for baseline survey, physical targets, monthly achievements, outcome monitoring framework indicators, groundwater/community outcomes and project implementation. The opportunity is to spatially connect these existing records instead of replacing them.

3.4 Scale mismatch

Field interventions can be small compared with 10–30 m satellite pixels. A check dam, recharge pit or small bund may not be directly detectable. The system should therefore support both intervention-level evidence and catchment/cluster-level outcome assessment.

3.5 Decision latency

A map that simply shows layers is not enough. The officer needs a ranked “what needs attention?” queue: missing evidence, inconsistent photos, strong negative change, weak post-intervention trend, data-quality failure, or outcome indicators moving below expected direction.

4. Existing Government Ecosystem to Design Around

The solution should fit the existing WDC-PMKSY 2.0 ecosystem rather than pretending to be a replacement. The official DoLR site states that WDC-PMKSY 2.0 is the current watershed-development component and explicitly encourages GIS and Remote Sensing use for project planning. The official MIS exposes state/district/project/year reports and outcome-monitoring structures.

Existing source / function	What JALDRISHTI should do with it
WDC-PMKSY 2.0 MIS	Ingest or emulate project, activity, work-ID, location and outcome records; preserve source IDs.
Baseline survey	Use as baseline context before interpreting change.
Physical Action Plan + Achievements	Compare planned vs completed interventions and spatially display gaps.
OOMF / outcome monitoring	Map outcome indicators to spatial units and time periods.
Geo-tagged photos	Turn each photo into a searchable evidence object with GPS, time, quality and intervention linkage.
SRISHTI-DRISHTI	Treat as a production satellite-data adapter; use public substitutes for prototype if required.
GIS/RS planning layers	Overlay land use, drainage, DEM-derived terrain, vegetation/water indices and intervention points.

5. Research Findings That Should Shape the Product

Finding 1 — GIS is explicitly part of the program workflow. The platform should look like an operational GIS tool with decisions and reports, not a consumer map.

Finding 2 — Outcome assessment is broader than imagery. Keep physical work completion, environmental indicators, water/soil/vegetation evidence and community/livelihood evidence in separate evidence groups.

Finding 3 — The official MIS already has outcome-oriented report structures. JALDRISHTI should complement those structures with spatial evidence and easier interpretation.

Finding 4 — Remote sensing is useful but imperfect. Clouds, seasonal variability, mixed pixels and small structures can produce misleading changes. Every metric needs data-quality status and confidence.

Finding 5 — The demo can be strongest at intervention level. A single well-explained intervention story can demonstrate the platform more convincingly than 30 decorative layers.

Finding 6 — AI should prioritize interpretation and triage. Use AI to summarize evidence, detect inconsistencies and prioritize field verification instead of fabricating precise ecological causal claims.

6. JALDRISHTI Solution Definition

JALDRISHTI = Jal + Drishti: a “water lens” over watershed intervention evidence. The platform has three connected layers: Spatial Context, Evidence Intelligence and Outcome Decision Support.

Layer	Modules	Judge-visible result
Spatial Context	Watershed boundary, drainage, terrain, land cover, water bodies, intervention points	Officer can see the project as a living spatial system.
Evidence Intelligence	Geo-photo ingestion, metadata validation, satellite time series, change detection, evidence linkage	Every intervention has an evidence trail.
Outcome Decision Support	Outcome scorecard, confidence, anomaly queue, AI summary, field-verification tasks, exportable report	The map leads to a clear action, not just visualization.

7. Core User Stories

- As a district officer, I can select a watershed and immediately see its interventions, current status and key outcomes.
- As a field officer, I can upload a geo-tagged photograph and link it to an existing work/intervention ID.
- As an analyst, I can compare pre- and post-intervention satellite indicators for a defined area and time window.
- As a reviewer, I can see why the system marked an intervention as improving, inconclusive or requiring verification.
- As a senior officer, I can export a one-page intervention evidence report with map, photo, trend and confidence.
- As a program manager, I can see where planned works exist without sufficient spatial evidence.

8. Evidence Model — The Main Differentiator

Use an evidence-bundle object for every intervention. The bundle is the atomic unit of JALDRISHTI.

Evidence field	Example
Intervention ID	WDC-GJ-RJK-014
Type	Check dam / farm pond / plantation / contour trench / recharge structure
Location	lat, lon, watershed ID, village, block, district
Field evidence	1–5 geo-tagged images, capture time, EXIF validity, image hash
Satellite evidence	sensor, date range, cloud mask, scene IDs, derived indices
Terrain context	elevation, slope, flow accumulation / drainage proximity
Outcome signals	NDVI delta, water persistence delta, soil-moisture proxy, LST where valid

Comparison design	before/after windows; optional nearby control/paired area
Evidence quality	Verified / Partial / Missing / Conflicted
Confidence	High / Medium / Low — based on data completeness, not “truth”
Human review	Pending / Accepted / Needs field verification
Audit trail	Source IDs, timestamps, transformation version

Important scientific rule

Do not label an intervention “successful” from one chart. Use “Observed positive change”, “Observed negative change”, “Inconclusive” or “Needs verification”, with a visible evidence explanation. This makes the demo more credible.

9. Geospatial Indicators and Analytics

Indicator	Formula / basis	Use in JALDRISHTI	Limit
NDVI	$(\text{NIR} - \text{Red})/(\text{NIR} + \text{Red})$	Vegetation response / recovery trend	Seasonality and crop cycles can dominate signal.
MNDWI	$(\text{Green} - \text{SWIR})/(\text{Green} + \text{SWIR})$	Open-water extent/persistence proxy	Small/temporary water bodies may be mixed pixels.
NDMI	$(\text{NIR} - \text{SWIR})/(\text{NIR} + \text{SWIR})$	Vegetation/canopy moisture proxy	Not a direct groundwater measurement.
LST	Satellite thermal product where valid	Thermal context around degraded/vegetated areas	Cloud/processing gaps; not a direct soil-moisture measure.
Slope / elevation	DEM-derived	Hydrologic and terrain context	Resolution strongly affects small interventions.
Drainage / flow accumulation	DEM processing	Context for drainage and intervention placement	Requires good DEM and hydrologic conditioning.
Change score	Normalized pre/post delta with QA gates	Summarize observed change	Prototype score; not a causal impact estimate.
Evidence completeness	Rule-based count of required evidence	Prioritize verification	Measures evidence availability, not project quality.

For the MVP, calculate indices in a precomputed local demo dataset rather than depending on live Earth Engine calls during the judging demo. The production architecture can later connect to Earth Engine, SRISHTI-DRISHTI or other approved geospatial services.

10. Data Strategy

Dataset class	Preferred source	MVP use	Production direction
Satellite optical	Sentinel-2 / Landsat 8/9	NDVI, MNDWI, NDMI, land-cover context	SRISHTI-DRISHTI + approved EO pipelines
Satellite thermal	Landsat 8/9 ST where valid	LST context	Validated thermal processing
DEM	SRTM/approved DEM	Elevation, slope, drainage/flow context	Higher-resolution approved DEM when available
Rainfall	NASA/IMD/approved precipitation source	Explain seasonal anomalies and pre/post windows	Official/approved meteorological API
Watershed boundaries	DoLR/Bhuvan/approved GIS datasets	Spatial unit of analysis	Official watershed boundary service
MIS/project data	WDC-PMKSY MIS	Works, achievements, outcome indicators	Official API/export/integration
Geo-coded photographs	Field capture	Intervention evidence	Mobile capture with EXIF/GPS validation

Public-data feasibility note: Sentinel-2 Level-2 collections are available through Google Earth Engine and its cloud-probability collection; Landsat 8/9 Collection 2 provides surface reflectance and land-surface-temperature products in Earth Engine. Landsat 8 and 9 observe the same location on an 8-day offset. These datasets make a prototype feasible even when the problem-specific platform is not directly accessible.

11. Competitive / Similar-Submission Scan

Public GitHub repositories already show teams attempting SIH26015 with generic dashboards, FastAPI stacks, React/Leaflet, remote-sensing analytics and evidence-oriented systems. This is important: a beautiful map alone will not differentiate the team.

Observed pattern in public submissions	JALDRISHTI response
Generic glassmorphism “mission control” dashboard	Avoid dashboard-only framing; lead with an intervention evidence workflow.
FastAPI + PostGIS + React/Leaflet architecture	Use a similar robust architecture if needed, but differentiate at the product/evidence layer.
Evidence reconciliation / audit-ledger approach	Borrow the principle of provenance and human adjudication; keep the MVP simpler.
Geospatial watershed intelligence platform	Make the demo story outcome-centric: “what changed at this intervention, and what should we verify?”

Competitive implication
The novel pitch is not “AI + GIS + satellite maps”. That combination is already visible in other public attempts. The stronger differentiator is an intervention evidence graph: field photo ↔ location ↔ work ID ↔ satellite time series ↔ terrain ↔ outcome indicators ↔ reviewer action.

12. Recommended Technical Architecture

Layer	Recommended technology	Reason
Frontend	React + TypeScript + Vite	Fast iteration, component ecosystem, easy Antigravity generation.
UI	Tailwind CSS + shadcn-style components / custom design system	Premium responsive UI without a heavy admin template.
Maps	MapLibre GL JS or Leaflet	Flexible geospatial rendering; no mandatory paid key for base prototype.
Charts	Recharts or ECharts	Fast time-series and indicator visualization.
Animation	Framer Motion	Polished transitions and evidence-drawer interactions.
Backend (optional MVP)	FastAPI	Python-friendly geospatial and analytics integration.
Database	PostgreSQL + PostGIS	Spatial queries and project/intervention relationships.
Storage	Object storage / Supabase Storage / local demo assets	Field photographs and generated reports.
Analytics	Python + rasterio/xarray/geopandas or Earth Engine	Raster processing and geospatial analysis.
AI layer	Pluggable LLM + deterministic feature extractor	Narrative explanation with reproducible base metrics.
Reports	HTML-to-PDF or Python report generation	Officer-ready evidence report.

13. Data Model (MVP)

Entity	Key fields
watersheds	id, name, district, state, geom, area_ha, baseline_date
projects	id, watershed_id, project_name, start_date, end_date, status, source_id
interventions	id, project_id, type, work_id, geom, planned_date, completion_date, status
field_evidence	id, intervention_id, image_url, capture_time, lat, lon, exif_valid, quality_score, hash
satellite_observations	id, intervention_id, sensor, date, scene_id, cloud_pct, ndvi, mndwi, ndmi, lst
outcome_assessments	id, intervention_id, before_window, after_window, metric_json, change_score, confidence, verdict
verification_tasks	id, intervention_id, priority, reason, assigned_to, status
audit_events	id, entity_type, entity_id, event_type, actor, ts, source, metadata

14. Complete Website / App Structure

Route / screen	Purpose	Must-have UI
/	Landing / pitch	Hero, problem, solution, evidence loop, mini map preview, CTA to demo.
/login	Role entry	Demo role selector + optional email login.
/dashboard	Command center	KPI cards, priority alerts, watershed map, trend chart, recent evidence.
/watersheds	Watershed explorer	Search/filter, map, watershed cards, status filters.
/watersheds/:id	Watershed detail	Map layers, intervention clusters, trend, outcome score, report CTA.
/interventions/:id	Evidence detail	Photo gallery, map pin, before/after charts, outcome panel, evidence lineage.
/field-capture	Field evidence	Photo upload, metadata validation, intervention link, GPS status.
/analysis	Temporal analytics	Date windows, index selector, change map, chart, compare mode.
/verification	Verification queue	Priority list, reasons, evidence completeness, reviewer actions.
/reports	Reports	Filter + generate/export intervention or watershed report.
/methodology	Methodology	Data sources, formulas, limitations, confidence model.
/settings	Demo/project settings	Role, data mode, thresholds, theme, about.

15. Page-by-Page UI Specification

15.1 Landing page

Light-first premium government-tech aesthetic. Full-width geospatial hero with subtle terrain/watershed contour background. Header: JALDRISHTI logo, Platform, Methodology, Demo. Hero headline: “From geo-tagged proof to measurable watershed outcomes.” Subheadline: “Connect field evidence, satellite change and watershed

intelligence in one auditable workflow.” Primary CTA: Open Command Center. Secondary CTA: View Evidence Story. Add a 3-step strip: Capture → Analyze → Verify.

15.2 Command center

Left navigation + top command bar. KPI cards: Active Watersheds, Interventions Tracked, Evidence Coverage, Verification Queue. Central map. Right side Priority Evidence Queue. Bottom trend chart. Map marker click opens evidence drawer, not a new page.

15.3 Watershed detail

Header with watershed name and status. Tabs: Overview, Interventions, Change, Evidence, Outcomes, Reports. Show map on left, selected intervention summary on right. “Outcome story” card shows observed change, evidence completeness and confidence.

15.4 Intervention evidence

This is the hero screen for judging. Large geo-photo, mini map, timeline, index cards, before/after swipe, evidence chain chips, AI narrative, limitations panel, and action buttons: Verify / Accept / Request More Evidence.

15.5 Analysis

Map plus time slider. Metric selector for NDVI, MNDWI, NDMI, LST. Before/after windows. Display a shaded change layer and a chart. Every visualization shows sensor, period and data quality.

15.6 Verification queue

Prioritized table/list with reasons: missing GPS, photo mismatch, weak post-trend, negative change, cloudy imagery, incomplete work record. Each item has “Review evidence”.

15.7 Methodology

Make scientific honesty a design feature. Include formulas, data sources, confidence logic and “What this system does not claim”.

15.8 Field capture

Mobile-first card layout with camera/upload, GPS status, work ID, photo quality, timestamp and submit. In demo mode, use simulated GPS and clearly label the mode.

16. Visual Design System for Antigravity

Token	Direction
Theme	Light government GIS / premium earth-observation product
Primary	Deep forest/teal; restrained use for controls and data status
Accent	Water blue; use for hydrology-related data, not for every element
Background	Warm off-white / very light gray with subtle contour/grid texture
Cards	12–18 px rounded corners, soft border, restrained shadow, high information density
Typography	Inter / Geist / system sans; large metric numbers; compact labels
Map	Neutral light basemap; semantic overlays for vegetation, water, risk, interventions
Status	Use shape + text in addition to color: Verified, Partial, Inconclusive, Needs Verification
Motion	200–300 ms transitions; drawer slide; map selection pulse; no

	excessive parallax
Responsive	Desktop first for officers, but all screens must work at 390 px mobile width

17. AI Layer — What to Build and What Not to Claim

Use a hybrid design. Deterministic geospatial metrics are the factual layer. AI is the interpretation and prioritization layer.

AI capability	MVP behavior
Evidence summarizer	Converts metrics + evidence metadata into a concise human-readable summary.
Anomaly detector	Flags unusual month-to-month or before/after patterns using deterministic thresholds/z-score-like rules.
Evidence completeness checker	Checks whether GPS, time, work ID, photo quality and satellite coverage exist.
Field-verification prioritizer	Ranks cases needing review based on evidence gaps / conflicting signals.
Photo classifier	Prototype only: classify broad intervention type with a pluggable model interface; no accuracy claim without labelled data.
Chat assistant	Optional: answer questions from the current watershed context and cite the evidence objects shown on screen.

Do not do this in the demo

Do not say “AI proved the check dam increased groundwater by 23%” unless the data and method support it. Say “Observed NDVI increased by X in the selected window; water persistence changed by Y; evidence confidence is medium; field verification is recommended.”

18. Prototype Outcome / Evidence Score

Use a transparent prototype index to give judges a single navigational summary. Label it clearly as a decision-support score, not an official DoLR program score.

Evidence_Readiness = 0.25 photo_quality + 0.20 geo_validity + 0.20 temporal_coverage + 0.15
work_record_match + 0.20 satellite_data_quality
Outcome_Signal = weighted, normalized directional changes in selected indicators after QA gates
Decision_Status = Positive_Signal / Negative_Signal / Inconclusive / Needs_Verification

The final UI should show the component contributions so the judge can understand why a case received its status.

19. Demo Data Plan (Critical for a 2-Day Build)

Object	Quantity	Purpose
Watersheds	3	Show filtering and comparison.
Projects	5	Show program hierarchy.
Interventions	18–30	Make the map feel operational.
Field photos	30–60 simulated/curated local assets	Demonstrate evidence bundles.
Satellite observations	~24 observations/intervention	Produce stable time-series charts.
Verification tasks	8–12	Populate review queue.
Reports	3 prebuilt HTML/PDF-like demo outputs	Show management workflow.

All synthetic demo values must be marked “Demo dataset” in the UI. Real public satellite examples can be substituted once data preparation is complete. Never present synthetic values as government observations.

20. 4-Minute Judge Demo Story

1. Open JALDRISHTI landing page and state the problem in one sentence: geo-tagged evidence exists, but it is difficult to interpret with spatial and temporal context.
2. Enter the Command Center. Click a watershed with a red/amber verification badge.
3. Click one intervention. The evidence drawer shows photo + GPS + work ID + satellite time series.
4. Switch from NDVI to MNDWI. Show the before/after period and change chart.
5. Open the Evidence Chain. Explain which facts came from field evidence and which came from satellite/terrain analysis.
6. Show the AI summary: “positive vegetation signal, inconclusive water signal, medium confidence”.
7. Click “Request Field Verification”. A task is created with the exact reason and missing evidence.
8. Generate a one-page report. Finish with the line: “JALDRISHTI turns spatial records into evidence-backed decisions.”

21. Build Sequence for Antigravity — Do This in Order

9. Create the React + TypeScript + Vite shell and install routing, Tailwind, Framer Motion, icons and chart/map libraries.
10. Implement the design system first: tokens, typography, buttons, cards, map legend, status chips, drawers and tables.
11. Create a local deterministic demo data module. Do not connect external APIs yet.
12. Build the Landing page and Command Center first.
13. Build the Watershed Explorer and Watershed Detail map interactions.
14. Build the Intervention Evidence page; this is the primary judging page.
15. Add Analysis page with local time-series data and metric switches.
16. Add Verification queue and action dialogs.
17. Add report generation/export using local data.
18. Add mobile field-capture screen.
19. Add methodology/limitations page.
20. Only after the full UI works, introduce backend adapters or live geospatial sources. Keep demo mode as the fallback.

22. Antigravity Master Prompt — Paste First

Build a production-quality React + TypeScript + Vite web application called JALDRISHTI for Smart India Hackathon 2026 problem statement SIH26015. JALDRISHTI means a water-focused “drishti”/lens for watershed intelligence. The product is an AI + GIS Watershed Intelligence & Field Evidence Platform for the Department of Land Resources / Ministry of Rural Development use case.

GOAL

Do not build a generic GIS dashboard. Build an evidence-first operational web platform that connects: watershed -> project -> intervention/work ID -> geo-tagged field photo -> satellite time series -> terrain context -> outcome indicators -> confidence -> field verification action.

DESIGN

Use a premium light government-tech visual system: warm off-white background, deep teal/forest primary, water-blue accents, clean typography (Inter/Geist), large whitespace, 14–18px rounded cards,

subtle borders and shadows, restrained glass effects, smooth 200-300ms motion. Responsive from 1440px desktop down to 390px mobile. No Bootstrap. No generic admin template.

TECH

React, TypeScript, Vite, Tailwind CSS, Framer Motion, Lucide icons, Recharts/ECharts, MapLibre GL JS or Leaflet. Use local deterministic mock/demo data first. No external API keys are required for the demo. Keep all data and analysis services behind typed interfaces so later we can replace mock data with FastAPI/PostGIS/Earth Engine/SRISHTI-DRISHTI adapters.

ROLES

Use a Demo Role switcher with: District Officer, Field Officer, Analyst. No complicated auth in the first iteration.

PAGES

- 1 / Landing
- 2 / login
- 3 / dashboard
- 4 / watersheds
- 5 / watersheds/:id
- 6 / interventions/:id
- 7 / field-capture
- 8 / analysis
- 9 / verification
- 10 / reports
- 11 / methodology
- 12 / settings

CORE INTERACTION

The key interaction is: select watershed -> select intervention marker -> open right-side evidence drawer -> see field photo, GPS/time metadata, work ID, before/after trend, index cards (NDVI, MNDWI, NDMI, LST where valid), terrain context, evidence completeness, confidence and AI summary -> choose Verify / Accept / Request More Evidence.

DATA MODEL

Create local typed objects for watershed, project, intervention, fieldEvidence, satelliteObservation, outcomeAssessment, verificationTask, auditEvent. Seed at least 3 watersheds, 5 projects, 24 interventions, 40 field evidence objects, 100+ satellite observations and 10 verification tasks. Mark all seed data as DEMO DATA.

ANALYTICS

Use deterministic prototype calculations. $NDVI = (NIR - RED) / (NIR + RED)$, $MNDWI = (GREEN - SWIR) / (GREEN + SWIR)$, $NDMI = (NIR - SWIR) / (NIR + SWIR)$. The UI can display precomputed demo values and trend deltas. Never claim causal impact. Verdict labels must be Positive Signal, Negative Signal, Inconclusive, Needs Verification. Show confidence and evidence quality separately.

AI

Create an "AI Insight Engine" service that receives an intervention evidence bundle and returns a structured summary using deterministic templates/rules. Keep the interface ready for a future LLM provider. Never fabricate sensor facts. AI text must reference the available metrics and explicitly mention limitations when confidence is not high.

DASHBOARD

KPI cards: Active Watersheds, Interventions Tracked, Evidence Coverage, Verification Queue.
Map center with layer control: Interventions, Water, Vegetation, Terrain, Verification.
Right panel: Priority Evidence Queue.
Bottom chart: monthly watershed health trend.

METHODOLOGY

Show formulas, data sources, confidence logic, and a prominent statement that remote sensing indicates observed change and does not by itself prove intervention causality.

QUALITY BAR

Use reusable components, empty/loading/error states, keyboard focus, tooltip explanations, accessible labels, consistent spacing, realistic microcopy, and polished responsive behavior. Avoid placeholder lorem ipsum. Use domain-appropriate copy everywhere. Make the product feel like a serious Indian government geospatial intelligence platform, not a student admin panel.

IMPORTANT

Keep the app runnable with `npm install + npm run dev`. Keep all mock data inside `src/data` or `src/mocks`. Put analysis logic inside `src/services/analytics`. Put UI in `src/components` and `src/pages`. Use React Router. Build the landing, dashboard and intervention evidence page first, then the remaining pages. Do not wait for backend APIs before making the UI complete.

23. Antigravity Follow-Up Prompts — Run One by One

Prompt 1 — Foundation

Refactor the current project into a clean production-style structure: `src/pages`, `src/components`, `src/layouts`, `src/data`, `src/services`, `src/types`, `src/hooks`, `src/lib`. Add strict TypeScript types and keep all demo data deterministic. Do not change the visual design unless necessary.

Prompt 2 — Design system

Create the JALDRISHTI light design system: warm off-white background, deep teal/forest primary, water-blue accents, neutral map palette, rounded cards, subtle borders, strong typography, status chips and map legends. Remove any generic dashboard/template styling.

Prompt 3 — Command center

Build the Command Center as the primary operational screen. Add KPI cards, interactive map, layer toggles, priority verification list, monthly trend chart and recent evidence. Selecting an intervention must open the evidence drawer without losing map context.

Prompt 4 — Evidence screen

Build the Intervention Evidence screen as the flagship SIH judging screen. Include hero geo-photo, metadata, mini-map, work ID, before/after charts, metric cards, evidence chain, AI summary, limitations and three action buttons. Make this page visually exceptional.

Prompt 5 — Analysis

Add the Analysis workspace with a time slider, metric selector, before/after windows, change layer, trend chart and data-quality banner. All values come from deterministic mock data. Show sensor and date ranges.

Prompt 6 — Verification

Build the Verification Queue with severity/priority, reasons, missing evidence, confidence, reviewer notes and action dialogs. Clicking an item should navigate to its evidence bundle.

Prompt 7 — Field capture

Build a mobile-first Field Capture flow. Show camera/upload placeholder, GPS status, capture time, intervention/work ID, image-quality status and submit flow. Label the demo GPS mode clearly.

Prompt 8 — Reports

Add report generation. Create a polished intervention evidence report card with map snapshot, photo, before/after trend, key indicators, evidence quality, confidence and reviewer status. Add print/download behavior.

Prompt 9 — Methodology

Create a methodology page explaining NDVI, MNDWI, NDMI, terrain layers, confidence and evidence limitations. Add a “What JALDRISHTI does not claim” section.

Prompt 10 — Polish

Perform a full UX polish pass. Fix spacing, typography, mobile layouts, map popovers, hover states, transitions, empty states and error states. Remove duplicate components and console warnings. Confirm npm run build succeeds.

24. Future Backend/API Contract

Endpoint	Purpose
GET /api/watersheds	List watersheds with summary KPIs.
GET /api/watersheds/{id}	Watershed details + intervention list.
GET /api/interventions/{id}	Full evidence bundle.
GET /api/interventions/{id}/observations	Satellite/indicator time series.
POST /api/evidence	Create/upload field evidence.
POST /api/assessments	Create/recompute outcome assessment.
GET /api/verification	Verification queue.
POST /api/verification/{id}/action	Accept / verify / request-more-evidence.
POST /api/reports/intervention/{id}	Generate report.
GET /api/methodology	Return data source and model metadata.

25. Risks and Mitigations

Risk	Impact	Mitigation
No direct SRISHTI-DRISHTI access during hackathon	High	Prototype with public Sentinel/Landsat + architecture adapter; say clearly what is production integration.
No central geo-tagged photo corpus	High	Use curated/simulated demo evidence and provide field-capture design; never present synthetic photos as government data.
Small intervention below satellite resolution	High	Cluster/catchment analysis + verification gate; never force per-structure satellite claims.
Clouds / seasonality	Medium	Cloud masking, season-matched windows, data-quality banners and inconclusive status.
Over-claiming AI	High	Deterministic base metrics, transparent AI summaries, no causal claims.
Overbuilding backend	High	Frontend-first deterministic demo; backend only after core UI is complete.
Generic dashboard appearance	High	Lead with intervention evidence chain and decision workflow.
Demo API/network failure	Medium	All primary judging flows must work offline from local demo data.

26. What We Should Produce After the Website

- Working web demo with one polished evidence story.
- Architecture diagram: Field → Spatial Join → Satellite/DEM → Analytics → Evidence Bundle → Decision.

- Data-flow diagram with public-data prototype and SRISHTI-DRISHTI/DoLR production adapters.
- One-page impact/evidence report screenshot.
- PPT for SIH submission/presentation using the same narrative.
- Demo script and 2–4 minute screen recording.
- README with setup, data provenance, limitations and future integration plan.

27. Research Sources Used

SIH26015 indexed current snapshot — title, metadata, 178/500 count and 5 Oct 2026 deadline

<https://sih2026-ps-viewer.vercel.app/ps/SIH26015>

SIH26015 problem statement mirror with full statement text and dataset pointer

<https://sih.iqubekct.ac.in/problems/SIH26015/>

Department of Land Resources — WDC-PMKSY 2.0 overview

<https://wdcpmkysy.dolr.gov.in/>

Department of Land Resources — watershed development documents

<https://dolr.gov.in/en/document-category/watershed-development-component/>

WDC-PMKSY MIS — reports and OOMF/report structure

<https://www.wdcpmkysy.dolr.gov.in/reports?roleId=1>

WDC-PMKSY MIS — monthly achievement / work report

<https://www.wdcpmkysy.dolr.gov.in/assetReport>

WDC-PMKSY 2.0 guidelines — GIS/RS planning, monitoring and evaluation section

https://www.wdcpmkysy.dolr.gov.in/reference/WDCPMKSY2.0_Guidelines.pdf

WDC-PMKSY user manual — baseline, plan/achievement, outcome updates

<https://wdcpmkysy.dolr.gov.in/reference/UserManual.pdf>

Google Earth Engine — Sentinel-2 cloud probability / 5-day revisit information

https://developers.google.com/earth-engine/datasets/catalog/COPERNICUS_S2_CLOUD_PROBABILITY

Google Earth Engine — Landsat 8 Level 2 dataset

https://developers.google.com/earth-engine/datasets/catalog/LANDSAT_LC08_C02_T1_L2

Google Earth Engine — Landsat 9 Level 2 dataset

https://developers.google.com/earth-engine/datasets/catalog/LANDSAT_LC09_C02_T1_L2

USGS — Landsat 8/9 acquisition schedule and 8-day combined offset

<https://www.usgs.gov/faqs/what-are-acquisition-schedules-landsat-satellites>

Public SIH26015 competitor example — PRAMAAN evidence reconciliation

<https://github.com/archdex-art/pramaan-sih26015>

Public SIH26015 competitor example — geospatial watershed intelligence platform

<https://github.com/M-Kishore92/Walkouts035-SIH26015>

Public SIH26015 competitor example — FastAPI/dashboard implementation

<https://github.com/xarjunpatil/SIH26015-Application-of-Geospatial-Techniques-for-visualization-and-analysis-to>

Research date: 3 October 2026. Current-count statements are explicitly time-stamped because submission counters can change quickly and public mirrors can disagree.

28. Final Build Decision

Build this first

A light-theme, map-first JALDRISHTI website whose hero flow is: Select watershed → select intervention → inspect geo-tagged evidence → compare satellite indicators → see transparent confidence → create field-verification task → export evidence report.

Do not spend the remaining build window on complex authentication, a huge backend, dozens of map layers, or unverified AI claims. The value of this concept is the quality of the evidence workflow and the clarity with which it shows how watershed monitoring can move from documentation to decision support.